

NIKOLAOS DIMITRIADIS

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nik-dim

Greece

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SUMMARY

I am a student researcher in Google DeepMind, working on continual learning and model merging with foundation models. I completed my PhD student at EPFL under the supervision of François Fleuret and Pascal Frossard where I focused on understanding the weight space and loss landscape of neural networks, such as Large Language Models (LLMs) and Generative models, and leverage these insights to develop efficient algorithms for model merging, task arithmetic and multi-task learning. I have also completed an internship in Google DeepMind, where I worked on editing text-to-image generative models to obtain new capabilities.

PROFESSIONAL EXPERIENCE

Google DeepMind

Student Researcher

Amsterdam, Netherlands

Sept 2025 – Present

- Conducting research on continual learning and model merging with foundation models.

Google DeepMind

Student Researcher

Zurich, Switzerland / Remote

Dec 2023 – Dec 2024

- Conducted research on text-to-image generative models.
- On site and full-time Dec 2023 - Feb 2024, remote and part-time Mar - Dec.

École Polytechnique Fédérale de Lausanne (EPFL)

Doctoral Assistant

Lausanne, Switzerland

Oct 2020 – Sept 2025

- Specialized in multi-task learning and model merging. My research led to state-of-the-art algorithms for efficient and continuous parametrization of the Pareto Front, and improved the compression and downstream performance when merging large pre-trained models post training.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL)

PhD in Computer Science

Lausanne, Switzerland

Oct 2020 – Sept 2025

- Research focus: Model Merging, Multi-Task Learning and Pareto Front Learning
- Advisors: François Fleuret & Pascal Frossard

National Technical University of Athens (NTUA)

BSc & MSc in Electrical and Computer Engineering

Athens, Greece

Sept 2014 – Sept 2020

- Thesis: “Tropical Geometry and Applications in Machine Learning and Optimization”
- Advisor: Petros Maragos

PUBLICATIONS

[1] Wang*, Qin*, Dimitriadis, Favero, Frossard. *MEMOIR: Lifelong Model Editing with Minimal Overwrite and Informed Retention for LLMs.* to be presented at NeurIPS (2025)

- [2] Garcia Giraldo, **Dimitriadis**, Wang, Frossard. *Single-Input Multi-Output Model Merging: Leveraging Foundation Models for Dense Multi-Task Learning*. arXiv (2025)
- [3] Wang*, **Dimitriadis***, Favero, Ortiz-Jimenez, Fleuret, Frossard. *LiNeS: Post-training Layer Scaling Prevents Forgetting and Enhances Model Merging*. ICLR (2025) [paper](#) [website](#)
- [4] **Dimitriadis**, Frossard, Fleuret. *Pareto Low-Rank Adapters: Efficient Multi-Task Learning with Preferences*. ICLR (2025)
- [5] Wang*, **Dimitriadis***, Ortiz-Jimenez, Fleuret, Frossard. *Localizing Task Information for Improved Model Merging and Compression*. ICML (2024) [paper](#) [website](#)
- [6] Matoba, **Dimitriadis**, Fleuret. *Benefits of Max Pooling in Neural Networks: Theoretical and Experimental Evidence*. TMLR (2023)
- [7] Yüzügüler, **Dimitriadis**, Frossard. *Flexible Channel Dimensions for Differentiable Architecture Search*. arXiv (2023)
- [8] **Dimitriadis**, Frossard, Fleuret. *Pareto Manifold Learning: Tackling Multiple Tasks via Ensembles of Single-Task Models*. ICML (2023)
- [9] **Dimitriadis**, Fleuret, Frossard. *SequeL: A Continual Learning Library in PyTorch and JAX*. CVPR Workshop on Continual Learning (2023) [website](#)
- [10] Yüzügüler, **Dimitriadis**, Frossard. *U-Boost NAS: Utilization-Boosted Differentiable Neural Architecture Search*. ECCV (2022)
- [11] **Dimitriadis**, Maragos. *Advances in Morphological Neural Networks: Training, Pruning and Enforcing Shape Constraints*. ICASSP (2021)

AWARDS

- **SNF/ACG Study Abroad Program Scholarship** 📅 Jul 2019
Awarded for summer school studies at the Imperial College Business School by Stavros Niarchos Foundation (SNF) and American College of Greece (ACG).
- **ACG Merit Scholarship & SNF Scholarship** 📅 Sept 2018 - Jun 2020
Dual scholarship for parallel studies at the American College of Greece.
- **Right to Virtuosity Distinction** 📅 Jun 2018
Awarded for extraordinary performance at the Classical Guitar Degree examinations of the National Conservatory of Greece.
- **Classical Guitar Degree** 📅 Jun 2018
Achieved after twelve years of studies at the National Conservatory of Greece.
- **"The Great Moment for Education" Eurobank EFG Scholarship** 📅 Oct 2014
Achieved the highest score at the national exams among students of "Ecole Franco-Hellénique Saint Joseph".
- **Sœur Sébastienne Scholarship** 📅 Oct 2007 - Jun 2014
Full academic scholarship for Junior High and High School at "Ecole Franco-Hellénique Saint Joseph".

TEACHING

- Courses: Deep Learning, Fundamentals of inference and learning, Probability and Statistics
- Co-advised multiple MSc theses with Swisscom and multiple MSc semester projects

SKILLS

Python Git PyTorch JAX Docker

LANGUAGES

English (fluent), French (intermediate), Greek (native)